

CHAPTER 10

STATISTICAL INVESTIGATION OF miRNA ALTERATION IN HYPERTHYROIDISM PATIENTS IN AL-ANBAR PROVINCE

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INTRODUCTION

Recently, there has been a great deal of interest in microRNAs, often referred to as miRNAs. MicroRNAs are non-protein-coding RNA molecules that are found in certain organs and are responsible for controlling gene expression in eukaryotic animals (Bavelloni et al. 2017). These microRNA molecules are tiny and their altered levels are diagnostic or causative in a number of pathological conditions, such as cancer or cardiovascular disease. miRNAs are an example of a promising new class of biomarkers that have the potential to be used in the diagnosis of life-threatening diseases as well as in tracking disease progression in patients. A significant number of the therapeutically targetable microRNAs are involved in the process of immune response control. Since their reduction has an impact on the overactive response to infection, this suggests that control of the immune response, which is also important in the regulation of genes, is crucial. This highlights the importance of regulating gene expression (Brosnan and Mitter, 2021).

The thyroid is a butterfly-shaped gland located in the centre of the neck, above the collarbones and below the larynx (voice box). Triiodothyronine (T3) and thyroxine (T4), two hormones produced by the thyroid, control how the body uses and stores energy (Beynon and Pinneri, 2016). In people with hyperthyroidism, the immune system produces an antibody similar to TSH, which causes the thyroid to overproduce thyroid hormone. This can happen to men and women of any age, although it most commonly affects women between the ages of 20 and 40. The symptoms of hyperthyroidism are caused by the thyroid gland becoming enlarged and producing excessive amounts of thyroid hormone (Stathatos, 2012). For therapeutic management of hyperthyroidism, early and accurate biomarkers to predict recurrence of hyperthyroidism are therefore of paramount importance. The role of microRNAs (miRNAs), a group of endogenous non-coding short RNAs, has recently received increased attention. In humans, miRNAs are located in the intron region of coding genes. These miRNAs can affect gene expression by acting on the promoter region of the target gene, preventing the binding of suppressor genes and inhibitory factors, thereby increasing gene expression. The hypermetabolic condition of

thyrotoxicosis is the end result of a group of illnesses known collectively as hyperthyroidism (Qadir and Faheem 2017). These conditions are characterised by abnormally high levels of thyroid hormones (T3 and T4) produced and released by the thyroid gland (Wilczynska and Bushell, 2015).

Both underactive and overactive thyroids may have a negative impact on fertility. However, there is little evidence to support the hypothesis that hyperthyroidism is associated with infertility, and this data is sometimes contradictory. Thyroid hormone has a number of effects on human reproduction, both at central and peripheral levels, and these effects are caused by different processes. It is possible for men and women with hyperthyroidism to experience infertility; however, this condition is often reversible once euthyroidism is achieved (Paraskevopoulou and Hatzigeorgiou, 2016). Certain microRNAs, which can be targeted therapeutically, play a crucial role in modulating the immune response. Their depletion has an effect on the hyperactive response to infection, demonstrating that regulation of the immune response, which also plays an important role in regulating gene expression, is essential (Ludwig et al. 2016). In the light of this information, the aim of this study is to investigate the relationship between thyroid antibodies (antithyroid peroxidase and antithyroid) and thyroid hormones and microRNA (miRNA) gene expression with hyperthyroidism in a specific group of women with hyperthyroidism.

MicroRNAs

MicroRNAs, also known as miRNAs, are short sequences of endogenous RNA that act as regulatory molecules but do not code for proteins. They have been found to play an important role in the regulation of gene expression in the majority of eukaryotic organisms. These naturally occurring RNA sequences are the focus of a considerable amount of research in a wide range of model species, from plants to mammals (Turchinovich et al. 2016). MicroRNAs are involved in a wide range of key biological activities in plants, including the control of growth and development and the response to abiotic stress. There is increasing evidence that microRNAs, which are found

in a wide range of animals, influence gene expression at the post-transcriptional level.

In the nucleus, stem-loop structures of about 70 nucleotides in length, called pre-miRNAs, are processed into mature microRNAs of 21 to 24 nucleotides in length. MicroRNAs are 21-24 nucleotides in length. Base pairings determined by sequence homologies are responsible for the formation of the stem and loop structures. Since the Pri-miRNA is thought to be longer than the pre-conserved miRNA stem-loop structures, the subsequent processing step involves some modification. In addition, microRNAs have been shown to stimulate gene expression under certain circumstances. Recent research has led scientists to hypothesize that microRNAs are transported from one subcellular compartment to another to modulate the rate of translation and perhaps even transcription. MicroRNAs are critical for proper animal development and play important roles in a variety of biological processes. In humans, abnormal expression of miRNAs has been linked to a wide range of diseases. In addition, microRNAs can be found in extracellular fluids, where they are released. Extracellular miRNAs act as signalling molecules that mediate interactions between cells (McGuire et al. 2015).

Biogenesis

The first step in miRNA synthesis involves the processing of RNA polymerase II/III transcripts, either following or simultaneously with transcription. MicroRNAs can occasionally be produced in clusters, which are large single transcripts. Since their seed regions are likely to be relatively similar, the miRNAs are then classified as belonging to the same family. There are two types of miRNA biogenesis pathways: canonical pathways and non-canonical pathways. Among these, the canonical biogenesis pathway is by far the most dominant and important miRNA processing pathway (Lukasik et al. 2016). The RNase III endonuclease Dicer is responsible for processing pre-miRNAs after they have been produced and subsequently exported to the cytoplasm by a complex consisting of exportin 5 (XPO5) and RanGTP. As a result of this processing step, a mature miRNA duplex is

formed, in which the terminal loop is removed. The mature version of the miRNA is named because the miRNA strand can be read in either orientation. These variations can range from almost equal amounts to predominantly one or the other (Mishra et al. 2016).

The degree of thermodynamic stability at the 5'-ends of the miRNA duplex and the presence of a 5'-U at nucleotide position 1 play an important role in determining whether the 5p or 3p strand is used. The passenger strands of the miRNA (Figure 1), which do not contain any mismatches, are selectively cleaved by AGO2 and subsequently degraded by cellular machinery, leading to a potential bias towards one strand. This happens when the miRNA duplexes have a central mismatch or lack AGO2-loaded miRNA (Lekchnov et al. 2016). After processing, the miRNA is incorporated into the RISC complex (RNA-induced silencing complex) to prevent translation of the target mRNA or to cleave the target mRNA. The AGO1 protein, which contains a PAZ domain, an RNA binding domain and an RNaseH-like PIWI domain, is a critical component of the RISC complex responsible for degrading miRNA* strands. The AGO protein family is vital and indispensable for the functionality of the miRNA-RISC complex, as its absence would render the complex ineffective (Militello et al. 2017).

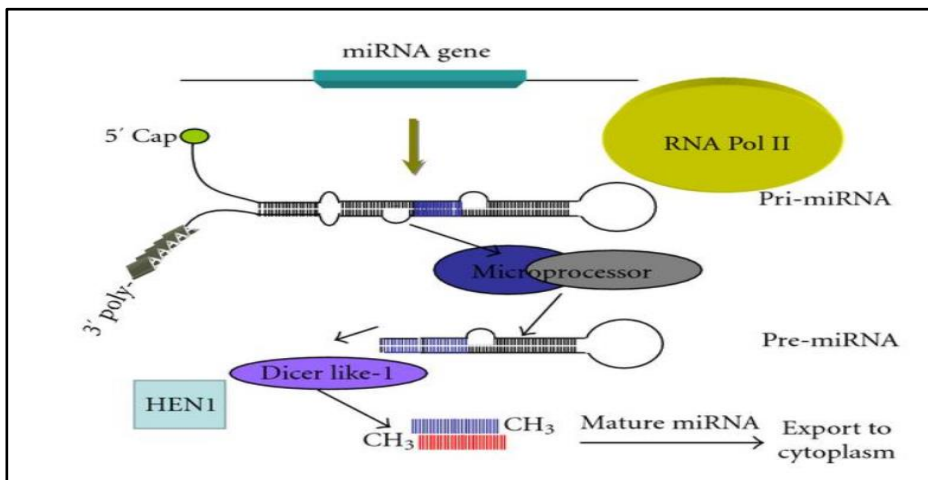


Figure 1. MicroRNA biogenesis (Achkar et al. 2016)

Mechanisms

Upon binding to specific sequences in the 3' untranslated region (UTR) of their target mRNAs, microRNAs initiate translational repression, mRNA deadenylation and decapping (Dori and Bicciato, 2019), as shown in Figure 2. Numerous research studies have consistently demonstrated the validity of this observation. In addition to the 5' ends of miRNA duplexes, miRNA binding sites have also been identified in different regions of the mRNA, including the 5' untranslated region, the coding sequence and even within promoters. However, further research is needed to fully understand the practical significance of this type of interaction (Michlewski and Cáceres, 2019). AGO and the guide strand form the miRISC, which is an acronym for the minimum miRNA-induced silencing complex. The degree of complementarity between MREs and target mRNAs determines whether there is AGO2-dependent slicing of the target mRNA or miRISC-mediated translational inhibition and degradation of the target mRNA. The induction of AGO2 endonuclease activity and mRNA cleavage can be attributed to an interaction between fully complementary miRNA and MRE (Rupaimoole et al. 2016).

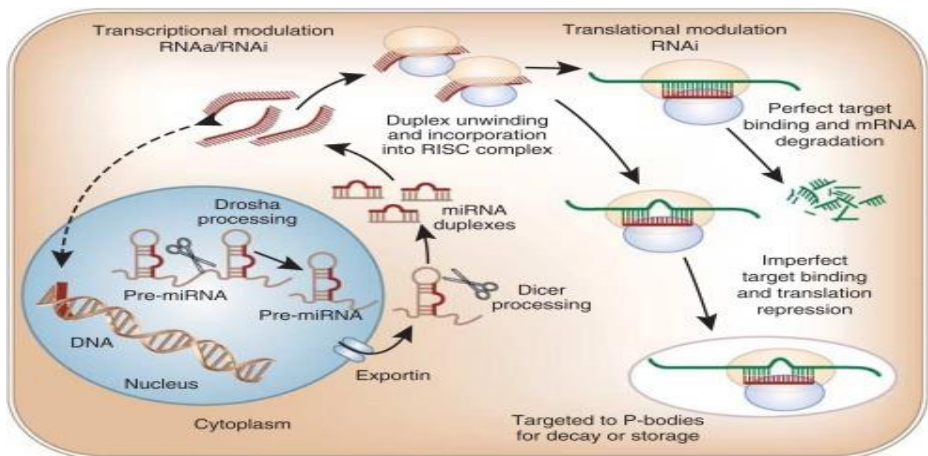


Figure 2. The role of miRNA (Cambiagno *et al.* 2021)

Hyperthyroidism

The condition known as hyperthyroidism, or an overactive thyroid, occurs when the gland responsible for thyroid function produces too much thyroid hormone. Excessive levels of thyroid hormones in the body lead to the acceleration of various physiological processes (Kravets, 2016). The condition known as hyperthyroidism (Figure 3) is characterized by a fast heartbeat, weight loss, increased hunger and feelings of worry. Surgery, anti-thyroid drugs, radioactive iodine and beta blockers are some of the treatments available for hyperthyroidism (Osuna et al. 2017).

In general, the thyroid gland is responsible for producing two primary hormones that affect how your body functions. These are called thyroxine (T4) and triiodothyronine (T3). Things like the rate at which your heart beats and the rate at which you burn calories are both controlled by your thyroid. It does this by releasing hormones that control your metabolism, the process by which your body converts food into energy and maintains its vitality (Srinivasan and Misra, 2015).

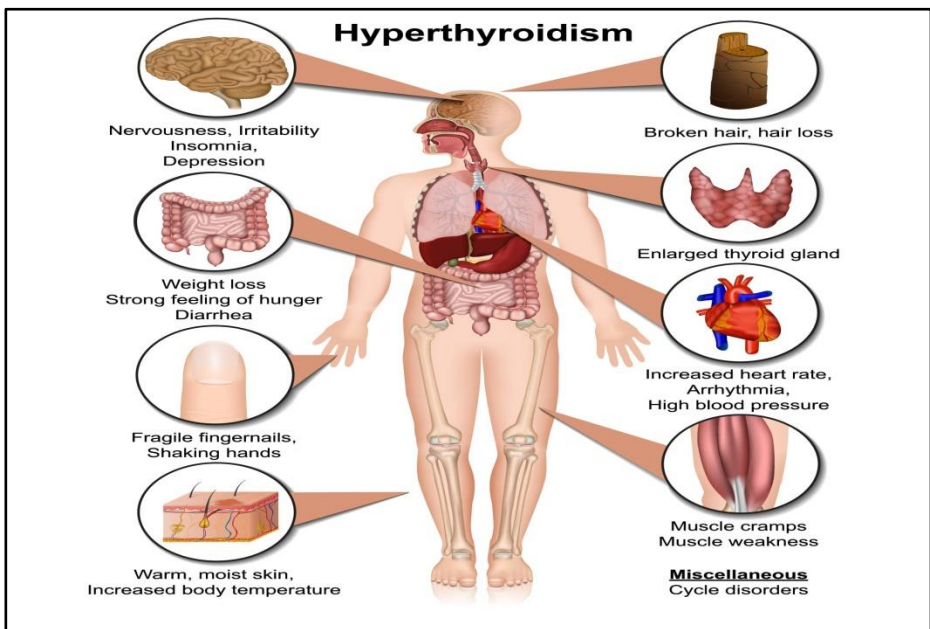


Figure 3. Overview of hyperthyroidism (Sjölin et al. 2019)

Hyperthyroidism can be caused by several conditions, including Graves' disease, Plummer's disease and thyroiditis. Thyroid hormones play an important role in regulating every aspect of the metabolic process. The thyroid plays an important role in producing a hormone called calcitonin, which helps maintain a healthy balance of calcium in the blood (Moleti et al. 2019).

The thyroid gland normally produces an appropriate amount of the hormone, but there are cases where it produces excessive amounts of T4. For example, Graves' disease is an autoimmune disease in which antibodies produced by the immune system cause the thyroid to overproduce T4, leading to symptoms of the disease. For unknown reasons, such as an autoimmune condition, your thyroid may become irritated after giving birth. Inflammation can cause excessive amounts of thyroid hormone, which should be kept in the gland, to be released into the bloodstream. Thyroiditis comes in different forms, some of which can be painful and some of which are not completely fatal.

Inflammation of the thyroid is called thyroiditis. Some forms of thyroiditis can cause thyroid hormone to leak from the thyroid gland into the bloodstream. This can be dangerous. As a direct result, you are at risk of developing symptoms of an overactive thyroid. Some forms of thyroiditis are associated with the development of hyperthyroidism. Subacute thyroiditis is characterized by a painful swollen thyroid gland enlarged by inflammation. Postpartum thyroiditis is an inflammation of the thyroid gland that can occur after a woman has given birth (Nguyen et al. 2018). Painless thyroiditis is a type of thyroiditis that can develop at any time and, unlike postpartum thyroiditis, is not caused by pregnancy. It is possible that your thyroid gland is too large. Experts agree that the cause of painless thyroiditis is most likely to be an autoimmune disease. Symptoms of hypothyroidism, commonly known as an underactive thyroid, can also be caused by thyroiditis. If your thyroid gland remains hyperactive for a long time, there is a chance that it will eventually become hypothyroid (Banigé et al. 2018).

Complications

If left untreated, hyperthyroidism can increase the risk of a thyroid storm. A thyroid storm is an extremely rare health condition that can develop if hyperthyroidism is left untreated for a long time. Because there is too much thyroid hormone in your system, your body goes into overdrive. A fast heart rate, increased blood pressure and fever are all symptoms that can occur during a thyroid storm, which is a potentially life-threatening combination of symptoms (Ferrari et al. 2019). Women who already have thyroid problems or who develop hyperthyroidism during pregnancy are both susceptible to developing pregnancy-related problems. Both the pregnant woman and the baby are at risk if there is an excess of thyroid hormone. There is a risk of ectopic pregnancy or premature birth. Having your thyroid hormone levels checked regularly during pregnancy can help detect any abnormalities, and your doctor may recommend treatment with medication (Andersen and Laurberg, 2016). If you have an overactive thyroid, your bones may become brittle and thin, putting you at risk of developing osteoporosis. Bone health can be improved by taking vitamin D and calcium supplements during and after treatment for the condition. Getting the recommended amount of exercise or doing any kind of physical activity regularly can also help prevent osteoporosis (Quérat et al. 2015). People with an overactive thyroid have an increased risk of developing thyroid cancer, often called thyroid cancer. According to a study summary published in 2018, hyperthyroid people with thyroid cancer had a more "aggressive" form of the disease, which was associated with a worse prognosis than euthyroid patients (those with a healthy thyroid). Atrial fibrillation is a potentially fatal arrhythmia (irregular heartbeat) that can be caused by hyperthyroidism. It can also cause congestive heart failure and increase the risk of stroke. Thyroid conditions that aren't properly managed can put a significant strain on your body and, if left untreated, can lead to life-threatening complications. There are a number of simple blood tests that can be used to diagnose hyperthyroidism and other thyroid problems (Bartalena et al. 2016).

Diagnosis

Because many of its symptoms are common to other conditions, hyperthyroidism cannot be diagnosed on the basis of symptoms alone. As well as ordering imaging tests, doctors may also order a range of thyroid blood tests to help confirm the diagnosis and identify the underlying cause. As hyperthyroidism has been linked to various fertility problems, it is advisable for women who are having difficulty conceiving to have their thyroid levels checked (Kaplowitz and Vaidyanathan, 2020).

Hyperthyroidism is characterised by increased levels of the thyroid hormones T3 and T4, and decreased levels of thyroid-stimulating hormone (TSH). The following imaging tests can be used to diagnose hyperthyroidism (Figure 4) and to examine the thyroid gland, a test known as radioactive iodine uptake (RAIU). In this test, you take a small amount of radioactive iodine, also known as a radiotracer, by mouth to determine the percentage of this substance that is absorbed by your thyroid gland (Vaske et al. 2016).

After a period of time, usually between six and twenty-four hours, the healthcare professional will scan your neck with a device called a gamma probe to see how much of the radioactive iodine your thyroid has absorbed. If a significant amount of radioactive iodine has been taken up by your thyroid, this indicates that your thyroid gland is producing excessive amounts of thyroxine (T4). If this describes your condition, you almost certainly have Graves' disease or thyroid nodules (King et al. 2016).

The gamma camera will take these images while you are in the same position as during the RAIU. The radioactive material may cause some or your entire thyroid to appear 'bright' on the screen. Ultrasound is a diagnostic procedure that uses high-frequency sound waves to create images of your thyroid gland. It is a non-invasive procedure that allows your doctor to see your thyroid gland on a screen. This test can be used by your doctor to look for nodules in your thyroid (Goichot et al. 2016).

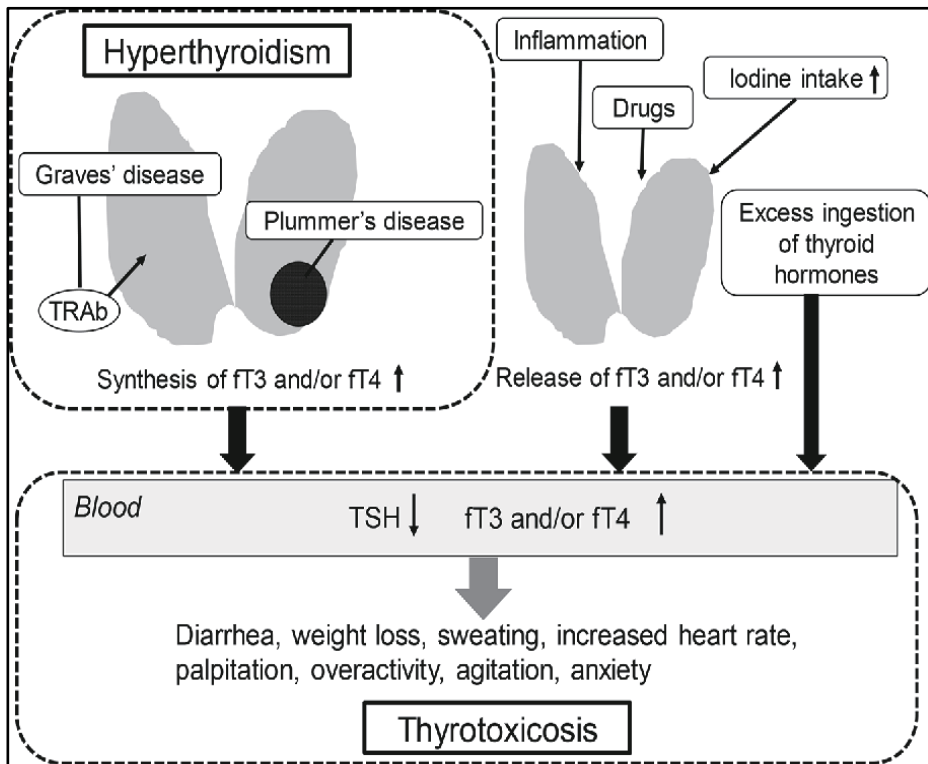


Figure 4. Diagnosis of hyperthyroidism (Watad et al. 2016)

Treatment

There are many different ways to treat hyperthyroidism. Depending on what caused your hyperthyroidism in the first place, some treatments may work better than others. Below are some of the treatments available for hyperthyroidism: Methimazole (Tapazol) or propylthiouracil (PTU), both anti-thyroid medicines: These medications suppress the production of thyroid hormone in the body, allowing for quick control of your thyroid condition (Donangelo and Suh, 2017).

Overactive thyroid cells can be treated with oral radioactive iodine therapy. Over the course of several weeks, radioactive iodine effectively eliminates these cells, resulting in a reduction in the size of your thyroid gland and lower thyroid hormone levels. This treatment usually results in permanent destruction of the thyroid gland and effectively treats hyperthyroidism. Unlike radioactive iodine uptake (RAIU) tests and diagnostic scans, the amount of radiation absorbed by

the body as a result of taking this medication varies (Dekkers et al. 2017). The majority of people who take this medicine will need to take thyroid hormone prescriptions for the rest of their lives to maintain normal hormone levels. These drugs, also called beta blockers, prevent the effects of thyroid hormone in the body. Although they do not change the actual levels of the hormone in your blood, these treatments can help relieve symptoms caused by hyperthyroidism, such as a fast heartbeat, anxiety and tremors. When treating hyperthyroidism over the course of a patient's life, this treatment is almost always combined with one or more other approaches (Turunen et al. 2020). The time taken to treat hyperthyroidism can vary greatly depending on the underlying cause of the condition. It may be treated with high doses of non-radioactive iodine drops. This treatment will restore thyroid levels within seven to ten days. However, this is only a temporary treatment and more permanent treatment, such as surgery, is likely to be needed in the long term. It is widely accepted as a long-term treatment option for hyperthyroidism (Léger and Carel, 2018). Most treatments come with a warning that they may have side effects. These potential risks include Methimazole and propylthiouracil are two drugs that can be used to treat hyperthyroidism (Gronich et al. 2020).

Both treatments have adverse effects (PTU). It is well known that these drugs can cause a wide range of adverse effects. PTU can cause liver damage, but only in a very small number of patients (less than 1% of the population). Agranulocytosis, a severe reduction in the number of white blood cells, is another harmful side effect that occurs in less than 1% of cases. All age groups are susceptible to these adverse effects (Sundaresh et al. 2017).

When taken by a pregnant woman, this medicine has the potential to cross the placenta into the developing foetus. This has the potential to cause hypothyroidism and goiter in the unborn child. Because of this potential adverse effect, pregnant women are closely monitored. There is also the possibility of an adverse reaction to these treatments, which occurs in about 5% of patients (Kobaly and Mandel, 2019).

With radiation, there is always a risk of developing cancer as an adverse effect. There is currently no evidence that treating

hyperthyroidism with radioactive iodine increases the risk of developing cancer. This risk is small and not very likely to occur. There is a known risk that radioactive iodine can be passed from a pregnant or breastfeeding mother to her baby. You should avoid using radioactive iodine while you are pregnant or breastfeeding as it may affect your child's thyroid. Loss of sensation in the mouth may occasionally occur after radioactive iodine (RAI) treatment (Scappaticcio et al. 2021).

Housekeeping Genes

Housekeeping genes are often described as essential, conserved, stably expressed in all cells and conditions, and members of cellular maintenance pathways. Broadly speaking, housekeeping genes are a set of genes that are essential for the maintenance of life. In practical terms, housekeeping genes can be defined as genes that show consistent expression in every cell of an organism, regardless of tissue type, developmental stage, cell cycle state or external signals. They serve as indicators of the biological well-being of an organism. These genes provide valuable information about species-specific and higher taxonomic level genomic features and gene functions that may influence conservation or evolutionary change. Consequently, a complete understanding of housekeeping genes can significantly improve exploratory, basic and translational research. However, despite their fundamental and translational importance, their definition has remained unchanged for more than 40 years (Bartha et al. 2018; Kiat et al. 2019).

Housekeeping genes have often been associated with the following four biological traits: evolutionary conservation, essentiality, involvement in cellular maintenance, and expression stability across samples. However, the link between these traits has not received much attention. The study of housekeeping genes is popular in both basic and translational science. Therefore, it is crucial to demonstrate the experimental basis for these standards (Joshi et al. 2022).

MATERIAL AND METHOD

General laboratory equipment, instruments and chemicals used during the study are as given in Table 1.

Table 1. Equipment and tools that were used

Equipment and instruments	Company	Country
Automatic micro pipette (100-1000µL)	Slamed	German
Automatic micropipette (2-20 µL)	Eppendrof	German
Centrifuge	Hattich	Germany
Refrigerator	Kelon	Turkey
Micro plate Elisa Reader	Rayto	China
Micro plate Elisa washer	Allsheng	China
VIDAS	Bio Merieux	France
Vortex shaker	INNOCOM	China
96-wall ELISA Plate	Bioneer	Korea
Vortex Mixer	TOPSCIEN	China

In Table 2, the kits used in the procedures of the study are given with their respective sources.

Table 2. The kits and their sources

Kit	Company	Origin
Thyroid stimulating hormone ELISA (TSH) kit	Bio merieux	France
VIDAS® FT4 kit	Bio merieux	France
VIDAS® FT3 kit	Bio merieux	France
VIDAS Anti-Tg ki	Bio merieux	France
VIDAS Anti-Tpo kit	Bio merieux	France
Anti-phospholipid ELISA kit	Aeskulisa	Germany

The trial will take place between 1 February 2022 and 1 July 2022. The study was approved by the ethics committees of the Department of Biology. A total of 165 women took part in the study. The experimental group consisted of 120 patient women (women with a history of miscarriage in the first trimester of pregnancy) and the control group consisted of 45 women. The women were aged between 19 and 46 years. Clinical information was collected directly from the patients. The aim of the study was to evaluate thyroid hormone and anti-thyroid hormone levels in the study group compared to the control group. The study also aimed to determine the percentage of thyroid disorders in women with a history of miscarriage in the first trimester

of pregnancy. Secondly, since pregnant women with a history of miscarriage in the first trimester are divided into normal weight and overweight women according to the average results, weight changes were analyzed to make comparisons between the two groups and to find out to what extent the likelihood of miscarriage is increased under the influence of weight changes.

A disposable plastic syringe was used to collect 5 mL of venous blood from pregnant women in three trimesters. The collected blood was then transferred to a gel tube and placed in a water bath where it was allowed to clot for ten minutes at 37°C. After clotting, the samples were centrifuged at 3000g for five minutes. The resulting clear serum was stored frozen in a refrigerator at -10°C. This serum was divided into separate portions for further analysis.

- Hormonal analysis.
- Immunological parameter assay.
- Molecular analysis.

Thyroid Function Tests (TFT) for TSH, FT4, FT3, ANTI-TPO and ANT-TG

Hormonal testing was performed using (VIDAS) from BioMerieux Company, France, by an enzyme linked fluorescence assay (ELFA).

Principle: The enzyme linked fluorescence immunoassay (ELFA) used in the VIDAS assay is performed on automated equipment. The instrument controls the temperature and each step of the assay. In this particular assay, the Solid Phase Receptacle (SPR®) acts not only as a pipetting tool but also as the solid phase, being a disposable alternative to a pipette. During manufacture, mouse anti-TSH, mouse anti-T4 and sheep monoclonal anti-T3 antibodies are coated onto the Solid Phase Receptacle (SPR) used for TSH analysis. If the sample contains Tg antibodies, these antibodies will form a specific bond with the protein coating the inside of the SPR. This binding occurs because the Tg antibodies interact specifically with the protein coating on the inner surface of the SPR.

The reagents required for the assay are contained in sealed reagent strips. A fluorescent substrate called 4-methylumbelliferyl phosphate is passed through the Solid Phase Receptor (SPR). The process that converts the substrate to the fluorescent product 4-methylumbelliferone is catalyzed by enzymes that are not released from the walls of the SPR. The concentration of TSH in the sample can be deduced from the result of the optical scanner's measurement of the intensity of the fluorescence, which is directly proportional to this concentration. The instrument automatically analyses and prints a report for each sample when the VIDAS, TSH, FT4, FT3, ANTI-TPO and ANTI-TG tests are completed. Automated analysis ensures efficient and accurate data processing, providing comprehensive reports that facilitate further evaluation and interpretation.

Determination of Thyroid Stimulating Hormone (TSH)

- Only what was needed was taken out of the package, and what wasn't used was returned to storage at 2-8 degrees Celsius as soon as possible.
- Approximately thirty minutes was allowed for the components to reach room temperature.
- One "TSH" strip and one "TSH" SPR were required for each test sample, control and calibrator. The pouch was then resealed for storage.
- The "TSH" code of the device serves as the test identification. The calibrator must be tested twice and marked 'S1'. The control should be marked C1 when tested.
- Mix the calibrator, control and samples (for serum or plasma separated from the pellet) using a vortex mixer.
- The total volume of the calibrator, control and test samples for the experiment is 200 liters.
- SPRs and "TSH" Reagent Strips were positioned in their usual locations on the instrument. SPRs and reagent strips with the assay code were compared for color consistency of the label.

- All assay steps were set up to be performed automatically by the instrument and the assay procedure was started according to the instructions in the user manual.
- The assay took approximately 40 minutes to complete and when it was finished the vials were resealed and returned to the correct temperature. The SPRs and strips were removed from the instrument after scanning.
- Used SPRs and strips were disposed of in the appropriate container. The normal range for micro international units per milliliter is 0.38 - 4.31 LU/mL.

Determination of the FT3 and FT4

- Once the necessary components had been removed from the kit, any other components that were not required were returned to storage at a temperature between 2 and 8 degrees Celsius.
- Allow approximately thirty minutes for the components to reach room temperature.
- After using one 'FT3' strip (or FT4 strip) and one 'FT3' SPR (or FT4 SPR) for each sample, control or calibrator tested, the storage bag was resealed. This was done to prevent contamination.
- The test was labeled on the instrument with the "FT3" (or FT4) code. "S1" was identified as the calibrator and was evaluated three times. In cases where the control was examined, it was given the designation "C1".
- A vortex type mixer specifically designed for use with serum or plasma after separation from the pellet was used to combine the calibrator, control and samples.
- During testing, the calibrator, control and sample test portions were measured at 100 μ L for FT3 and 200 μ L for FT4.
- The "FT3" SPRs (FT4 SPR) and "FT3" strips (FT4 strip) were inserted into the device. The color labels on the SPRs and Reagent Strips were checked to ensure they matched the corresponding assay code.

- The test procedure was started by following the instructions in the user manual. The instrument performed the entire procedure without human intervention.
- After pipetting, the vials were securely re-closed and stored at 2-8°C.
- The test was completed in just under half an hour. After analyzing the test results, the SPRs and strip cassettes were removed from the device.
- Normal ranges for FT3 = 2.17 - 3.34 pg/mL (pictograms per milliliter).
- Normal ranges for FT4 = 0.82 - 1.63 ng/dL (nanogram/deciliter).

Determination of anti-thyroglobulin antibody and anti-thyroid peroxidase

- The required reagents were taken from a ready-to-use refrigerator.
- Each specimen, control and calibrator was tested with an "ATG" Strip (ATPO Strip) and an "ATG" SPR® (ATPO SPR). Carefully reseal the storage pouch after removing the required SPRs.
- The "ATG" (ATPO) code on the instrument is used to identify the test being performed. The calibrator is marked "S1" and checked three times. If the control is attempted, it should be marked "C1".
- A vortex type mixer was used to combine the calibrator, control and samples.
- For this test, the calibrator, control and sample test portion is 100 µL for both ATPO and ATG.
- We loaded the instrument with an ATG, some SPRs (ATPO SPRs) and some ATG strips (ATPO strips). The color labels on the SPRs and reagent strips were checked to ensure they matched the assay code.
- The test was started after reading the manual recommendations. The instrument performed the entire assay procedure mechanically.

- After pipetting, the vials were reclosed and stored at 2-8°C.
- The test was completed in just over a quarter of an hour. At the end of the test, the SPRs and strips were removed from the device and the used SPRs and strips were disposed of in the designated container.
- Normal ranges for anti-thyroglobulin antibody (ANTI-TG) = 2.68 - 33.2 ng/mL.
- Normal ranges for anti-thyroid peroxidase (ANTI-TPO) < 8.0 IU/mL.

Determination of immunological parameters

Estimation of anti-phospholipid autoantibodies (APL) screen IgM was performed using ELISA kits according to the manual procedure provided by Aeskulisa Company (Germany). The analysis was performed in human serum using a microplate enzyme-linked immunosorbent assay for quantitative determination.

Principle: Serum samples are diluted 1:101 and incubated in microplates coated with a specific antigen. If the patient's antibodies are present in the sample, they will bind to the antigen. Any unbound fraction is removed by washing in the next step. Anti-human immunoglobulins conjugated to horseradish peroxidase (conjugate) are then added and incubated with the antigen-antibody complex in the microplates. In the next step, any unbound conjugate is washed away. The addition of TMB substrate then triggers an enzymatic colorimetric reaction characterized by a blue color. The addition of dilute acid stops this process and causes the color to change to yellow. The initial concentration of each antibody in the patient sample is directly proportional to the amount of conjugate bound to the antigen-antibody complex formed as a result of the reaction. This means that the amount of conjugate bound to the antigen-antibody combination is directly related to the degree to which the chromogenic contributes to the color intensity.

Molecular studies

Extraction of genomic RNA; RNA was extracted using a commercial TRIzol extraction kit following the steps of RNA

separation, RNA precipitation, RNA washing and RNA solubilisation. The RNA was reverse transcribed (RT) to complementary DNA (cDNA), which was used as a template in the PCR reaction. AccuPower Rocket Script RT PreMix from Bioneer/Korea was used for the RT reaction. This product is a lyophilised master mix ready for immediate use. This kit contains everything you need to synthesize first-strand cDNA from an RNA template, including all the essential components. The RT-PCR programme for complementary DNA synthesis was as given in Table 3.

Table 3. RT-PCR program to complementary DNA synthesis

Step	Temperature				Time
	dN6	dN12	dT20	Specific primer	
Primer annealing	15°C	30°C	37°C	Tm of primer	10min
cDNA synthesis			42°C		30min
Heat inactivation			95°C		5min

RT-PCR was used to detect gene expression of the microRNA gene. Species-specific micRNA146a primers were used to develop RT-PCR primers sensitive enough to detect gene expression of the blood micRNA gene. The primers were supplied by Macrogen Company as lyophilised products in different picomole concentrations. The primers were prepared according to Macrogen's protocol. Some of these details are shown in Table 4.

Table 4. RT-PCR master mix used to detect the micRNA146a gene expression

Component	Concentration	Amount (μL)
Master Mix	-	10
micRNA146a -F primer	10 μM/μL	1
micRNA146a -R primer	10 μM/μL	1
Nuclease free water	-	6
cDNA sample	-	2
Total	-	20

The following RT-PCR protocol was used to determine the expression of the micRNA146a gene (Table 5).

Table 5. RT-PCR program used to detect the micRNA146a genes expression

Step	Temperature	Time	Cycles
RT inactivation/Hot-start activation	95 °C	2 Minutes	1
Step qPCR a. Denaturation b.	95 °C	15 Seconds	40
Annealing and extension	72 °C	1 Minute	
	72 °C	30 Seconds	
Dissociation	95 °C	30 Second	

Statistical Analysis

The Statistical Analysis System (SAS) programme was used for data analysis to determine the effect of various factors on the study parameters. The mean and standard deviation were calculated for all parameters. After the Kruskal-Wallis test to determine whether the data were normally distributed, the Student t test was used to assess the significance of differences between normally distributed variables. A significance level of 0.05 was used for statistical comparisons.

RESULTS AND DISCUSSION

Serum hormone concentrations including FT3, FT4, TSH, ANTI-TPO and ANTI-TG were measured using the Vidas method. A comparison was made between the study and control groups and significance was assessed using analysis of variance (ANOVA) and t-test. The results are shown in Tables 6-7.

Table 6. Comparing the levels of TSH and thyroid hormones between expectant women in their first trimester and their controls

Parameters	First trimester (Mean $\pm$ SD) n= 120	Control (Mean $\pm$ SD) n= 45	P-value
TSH (μ LU/ML)	4.417 $\pm$ 0.613	1.889 $\pm$ 0.222	0.046*
FT3 (pg/mL)	3.092 $\pm$ 0.463	2.945 $\pm$ 0.603	0.628
FT4 (ng/dL)	1.735 $\pm$ 0.287	1.192 $\pm$ 0.296	0.391

The results showed that the TSH level in the first trimester was (4.417 $\pm$ 0.613) μ IU/mL, which was higher than the mean $\pm$ SD (1.889 $\pm$ 0.222) μ IU/mL observed in the control group. This study shows that there is a significant difference in TSH levels between the control and patient groups ($P < 0.05$). FT3 was (3.092 $\pm$ 0.463), lower than the mean $\pm$ SD (2.945 $\pm$ 0.603) pg/mL of the control group. This shows that there is no significant difference ($P = 0.628$) in FT3 levels between controls

and patients. FT4 is (1.735 ± 0.287), which is closer to the mean $\pm$ SD (1.192 ± 0.296) ng/dL of the control group. Furthermore, this study shows that there is no significant difference in FT4 levels between controls and patients ($P=0.391$).

Table 7. A comparison of APL and anti-thyroid hormone levels between first-trimester pregnant women and their control expectant counterparts

Parameters	First trimester (Mean $\pm$ SD) n= 120	Control (Mean $\pm$ SD) n= 45	P-value
APL (U/mL)	7.087 ± 0.8426	4.762 ± 0.3437	0.045*
ANTI-TPO (IU/mL)	83.665 ± 17.565	4.595 ± 0.379	0.013 **
ANTI-TG (ng/ml)	19.325 ± 1.9816	16.588 ± 0.9915	0.549

Table 7 shows the mean levels of anti-TPO, anti-TG and anti-phospholipid antibodies in the sera of pregnant women and control group patients. The mean $\pm$ SD value of ANTI-TG was (19.325 ± 1.9816) ng/mL, which is higher than the mean $\pm$ SD value of the control group (16.588 ± 0.9915) and not significant.

Moreover, the result shows high significance between ANTI-TPO (0.05) and APL (0.01) when compared with the control group. ANTI-TPO (83.665 ± 17.565) IU/mL and anti-phospholipid (7.087 ± 0.842) U/mL values were all higher than the control group.

Our study found that the risk of miscarriage was significantly higher in pregnant women who were diagnosed with autoimmune thyroid disease (AITD) in the first trimester of their pregnancy compared with those who were not diagnosed with AITD. This finding is consistent with the study by (Glinioer et al. 1994). Another research study conducted by (Triggianese et al. 2021) showed that pregnant women who experienced recurrent miscarriage had a prevalence of thyroid autoimmunity of (37.8%) compared to the control group, which was statistically higher ($P = 0.01$) than the control group.

According to the division of body mass of women in the target group for study (first trimester group) into normal weight (BMI = 18.5-24.9) and overweight (BMI = 25-29.9), there was a significant difference ($p \leq 0.001$) between the body mass index, where the mean $\pm$ SD body mass of pregnant women with normal weight is ($23.895 \pm$

2.454) and the mean $\pm$ SD body mass of overweight is (26.738 ± 3.964). Also the table shows the mean $\pm$ SD of abortion in overweight (1.900 ± 0.834) is high of normal weight that mean $\pm$ SD (1.815 ± 0.827) with no a significant as shown in Table 8.

Table 8. Mean and SD of abortion in women of normal weight and overweight

Groups		N	Mean $\pm$ SD	P – value
BMI	Normal weight	85	23.895 ± 2.454	0.000
	Over weight	35	26.738 ± 3.964	
Abortion	Normal weight	85	1.815 ± 0.827	0.524
	Over weight	35	1.900 ± 0.834	

Our research found that women with a higher than normal body mass index (BMI) were more likely to have a spontaneous abortion.

In addition, total RNA extraction was successfully performed from a variety of samples with RNA concentrations ranging from 62 to 288ng/ μ L. Aseptic procedures are required to obtain high concentrations of total RNA and are highly dependent on the extraction conditions. The use of TRIzol is recommended for total RNA extraction from blood samples.

After RT-PCR analysis and normalization to the expression of a housekeeping gene, we were able to find the MIR146A gene in these samples. The $2^{-\Delta\Delta C_t}$ approach comparing threshold cycles (CT) was used to quantify relative changes in expression levels. Amplification and detection of the MIR146A gene was performed using SYBR Green qRT-PCR.

Positive results in hyperthyroidism showed amplification at CT (threshold cycle) values in the range of 16.38-17.04 for the housekeeping gene and 28.86-29.42 for the MIR146A gene (Figure 5).

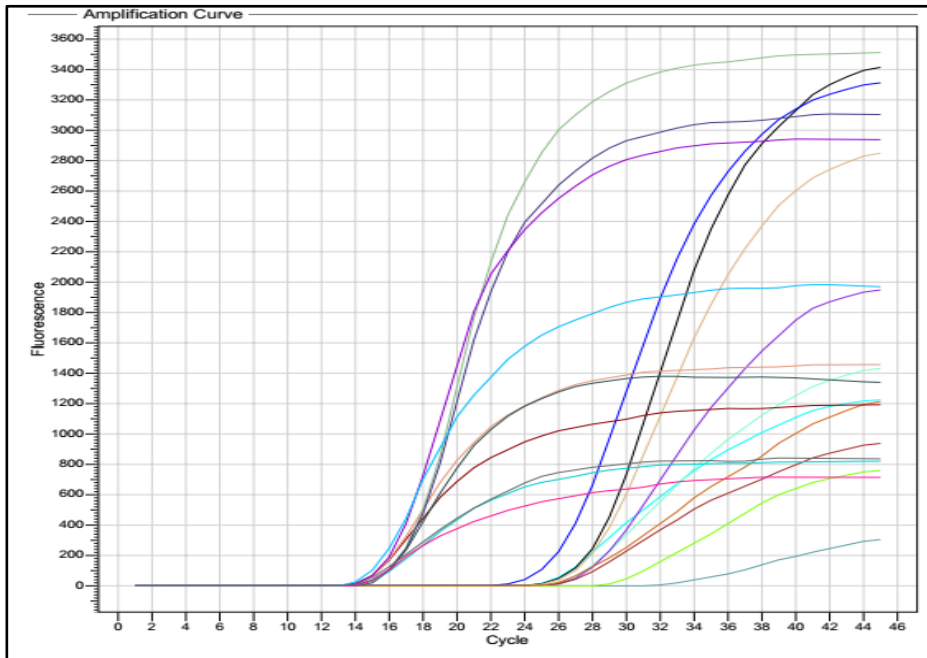


Figure 5. Curve of cycling of *MIR146A* gene and housekeeping gene

The melting temperature (T_m) values obtained for the hyperthyroidism patients and the control sample; the melting temperature was in the range of 81.64, 82.04, 81.94, 81.44, 81.04, 81.34, 82.54, 82.64, 82.34, 82.44 °C respectively for the housekeeping gene and the melting temperature of *MIR146A* was in the range of 81.24, 82.44, 54, 81.94, 81.24 °C (Figure 6).

The expression of microRNA by itself or in conjunction with multi-chaperone complexes acts as a significant repressor of heat shock transcription factor. Research by Zuehlke et al. (2015) elucidates that in some disease cases, increased expression of *MIR146A* could potentially serve as a prognostic indicator.

Furthermore, Calderwood (2018) reported that a significant upregulation of numerous microRNAs, including *MIR146A*, *MIR146B*, *MIR148A* and *miR-146a-5p*, was observed in several cancer types, such as breast, prostate, lung and melanoma. This increased expression of microRNAs is strongly associated with poor patient outcomes. MicroRNA itself is overexpressed in thyroid cancer, as observed in the study by Soudry et al. (2017). Furthermore,

increased microRNA expression is associated with poor patient survival. This provides a strong rationale for inhibiting microRNA function as an innovative therapeutic approach in the treatment of certain subtypes of thyroid cancer, as suggested by Lettini et al. (2020).

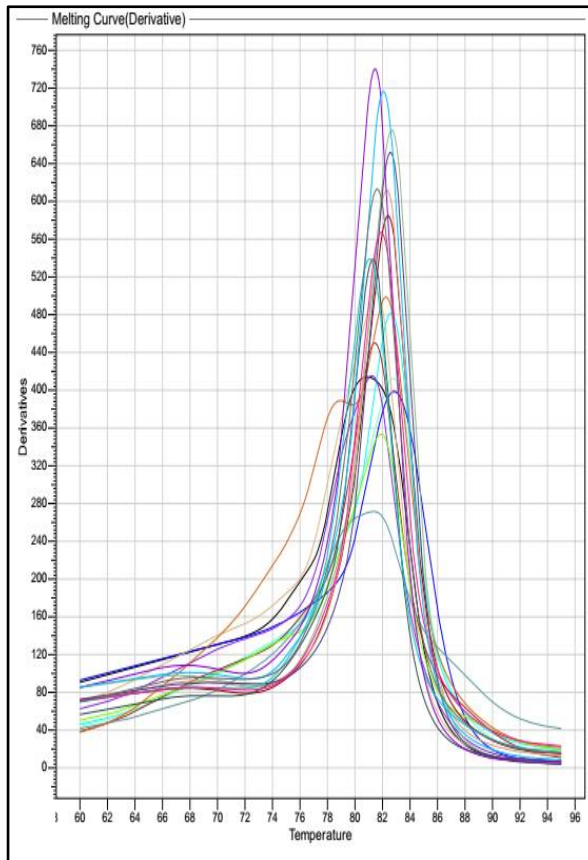


Figure 6. Curve of melting *micRNA146A* gene and housekeeping gene

CONCLUSION AND DISCUSSION

Thyroid hormones (FT3, FT4) and TSH are associated with oxidative stress and antioxidant status in patients with hypo- and hyperthyroidism. The *micRNA146A* genes can be used as a risk indicator in the diagnosis of thyroid disorders. Real-time PCR is an important tool to determine the gene expression of *micRNA146A* genes as risk factors, which could be a powerful molecular tool to study the relationship between these genes and thyroid disorders.

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